Happy Health – Unified Data Platform

Solution design · Recommended platform: Microsoft Fabric

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Summary: our approach and platform choice

Our approach in one paragraph

Build one data platform that keeps every source record exactly as it arrived, turns it into one standard model, and matches each patient across every system. Use an open-source storage format and modern analytics architecture that can be shared with multiple workloads. The main focus for the first 120 days is on the five CMS quality measures, because they need the hardest groundwork: patient matching, a clean model of visits and a complete audit trail. Every other group reuses that groundwork later.

Why Microsoft Fabric

- One platform for loading, storing, transforming, reporting and alerting. Allows to escalate and supplies on demand upgrade for peak workloads.
- Healthcare Data Solutions provides ready-made pipelines that put Epic and MEDITECH records into one standard format (FHIR)
- Built-in security and governance (Purview, Entra ID, row- and column-level access) to prove each person sees only what they need
- Power BI is part of the platform, so dashboards read the data directly and self-service reports stay safe
- Open table formats and open-source tools (Splink for matching, Presidio for de-identification) limit lock-in

Explanation

Our approach is to build one platform with three jobs. First, it keeps every record exactly as it arrived from Epic, MEDITECH and the ancillary systems, which gives us the audit trail. Second, it turns those records into one standard model, so a MEDITECH visit and an Epic visit look the same. Third, it matches each patient across all systems, because no shared patient ID exists today.

We spend the first 120 days on the five CMS (Centers for Medicare & Medicaid Services) quality measures because they keep the program funded and need the hardest foundations. Population health, self-service reporting and the research extract all reuse those foundations afterwards.

We recommend Microsoft Fabric for five reasons. It is a single platform covering loading, storage, data engineering, reporting and alerting, which matters on an incremental budget. Its Healthcare Data Solutions module provides tested pipelines for the FHIR healthcare standard. Security and governance are built in through Purview, Entra ID and OneLake security. Power BI is native, so dashboards read the data without copying it. And because tables are stored in an open format and we use open-source tools such as Splink and Presidio, moving later would mean moving storage, not redesigning.

Databricks and Snowflake were credible alternatives. Fabric wins here on having one platform, native Power BI and the healthcare module.

1. Architecture – the complete picture

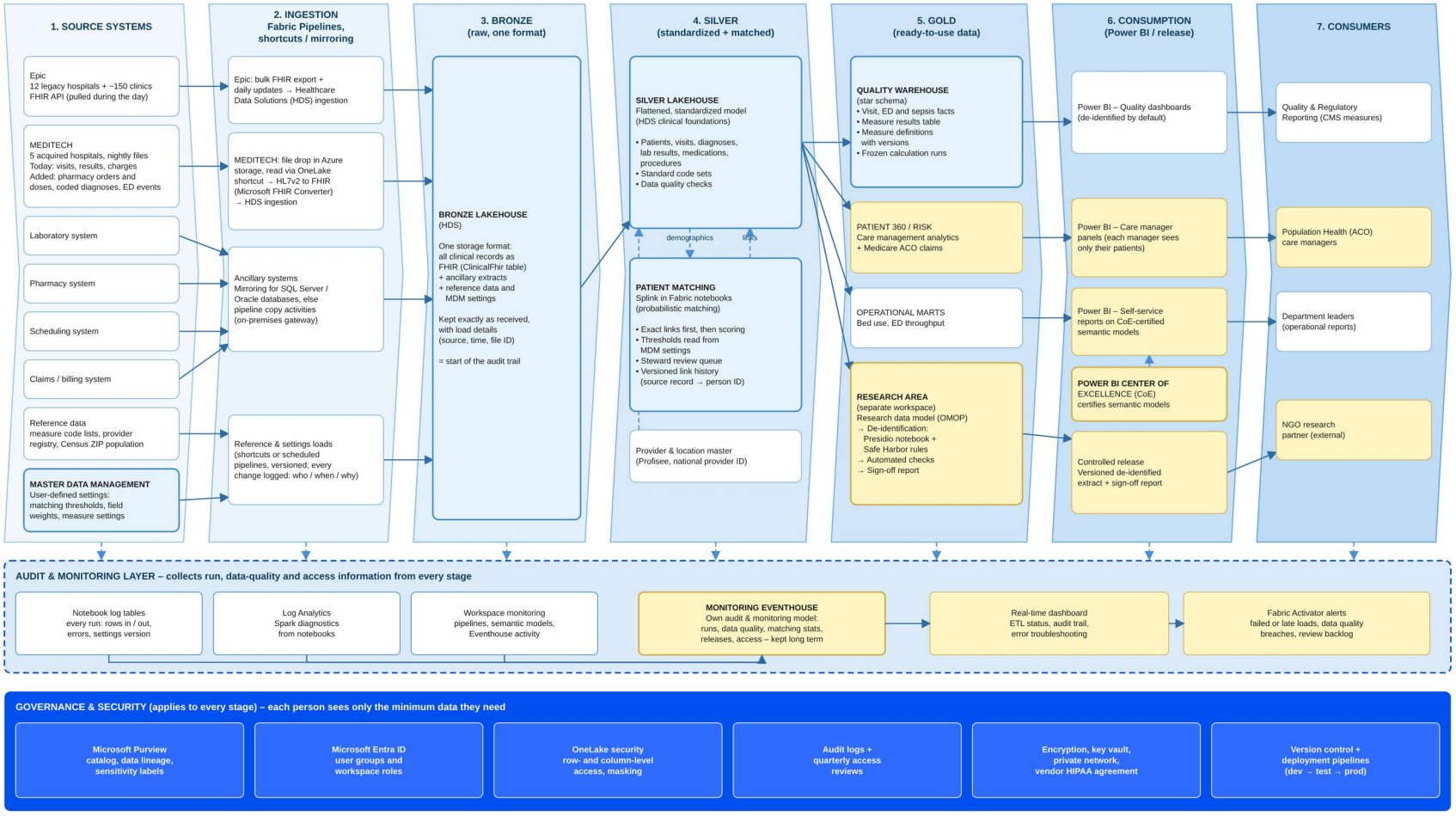


Figure 1. Architecture – the complete picture

Explanation

This diagram shows the target state. Components shaded pale yellow come after the 120-day go-live. Data flows left to right through seven stages, drawn as arrows to show the direction.

Stage 1, sources: Epic at the 12 legacy hospitals and about 150 clinics offers a live FHIR interface. The five MEDITECH hospitals send nightly HL7v2 files, which we assume will be extended with pharmacy, diagnosis and emergency department tracking messages. Four ancillary systems each keep their own patient ID. Reference data supplies measure code lists, and a new master data management (MDM) source holds the settings people control, such as matching thresholds.

Stage 2, ingestion: Fabric Pipelines schedule every load, OneLake shortcuts read data that already sits in cloud storage without copying it, and Mirroring keeps supported databases in sync. Healthcare Data Solutions ingests Epic's FHIR files directly. MEDITECH's HL7v2 messages are converted to FHIR first and then follow the same path.

Stage 3, bronze: every record lands in one storage format, kept exactly as received. This is the start of the audit trail.

Stage 4, silver: records are flattened into a standard model and checked for quality. Splink matches patients across systems using thresholds from the MDM settings.

Stage 5, gold: ready-to-use data for each group, including the quality warehouse and an isolated research area that de-identifies data with Presidio.

Stages 6 and 7: Power BI and a controlled release channel serve the four consumer groups.

Under the stages, the audit and monitoring layer collects run and error information from every stage into its own model, which feeds a real-time dashboard and alerts. The governance and security band applies everywhere.

1. Architecture – Stage 1: source systems

Eight sources, three ways of connecting, no shared patient ID

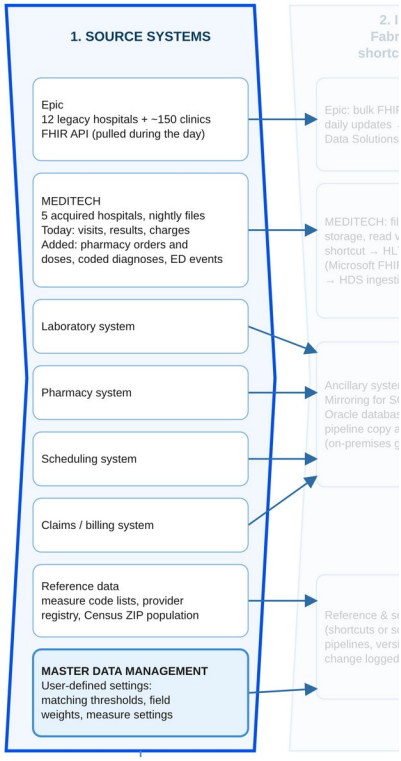


Figure 2. Stage 1: source systems highlighted in the architecture

Summary

- Epic (12 legacy hospitals + ~150 clinics): live FHIR interface for patients, visits, diagnoses, results, orders and procedures
- MEDITECH (5 acquired hospitals): nightly HL7v2 files, assumed to be extended with pharmacy orders and doses, coded diagnoses and emergency department tracking events
- Laboratory, pharmacy, scheduling and claims systems, each with its own patient ID
- Reference data (measure code lists, provider registry, Census ZIP populations) plus master data management settings such as matching thresholds

KEY DESIGN CHOICE
With the MEDITECH files extended, SEP-1 and emergency department time to provider can be calculated at all 17 hospitals.

Explanation

Stage 1 is the source landscape. The platform adapts to the sources rather than asking them to change, with one exception that we state as an assumption.

Epic covers the 12 legacy hospitals and about 150 clinics. Its live FHIR interface (FHIR is the modern healthcare data standard) gives us patients, visits, diagnoses, lab and vital-sign results, medication orders and procedures.

The five MEDITECH hospitals send nightly HL7v2 files, the older healthcare messaging standard. Today those files cover visits, results and charges only. SEP-1 needs the exact times antibiotics and fluids were given, and emergency department time to provider needs the time a provider first saw the patient. We therefore assume the MEDITECH nightly files are extended with three additions: pharmacy orders and doses given (RDE and RAS messages), coded diagnoses and procedures (DG1 and PR1 segments), and emergency department tracking events such as provider seen, room assigned and decision to admit.

The four ancillary systems each have their own patient ID. Claims uses the insurance subscriber ID, which often belongs to someone other than the patient.

Reference data includes the official measure code lists, the national provider registry and Census ZIP-code populations needed for de-identification. The master data management source holds settings that business users control, not code: matching thresholds, field weights and measure settings. Data stewards own them, and every change is versioned.

1. Architecture – Stage 2: ingestion (pipelines, shortcuts and mirroring)

One repeatable loading pattern per type of source

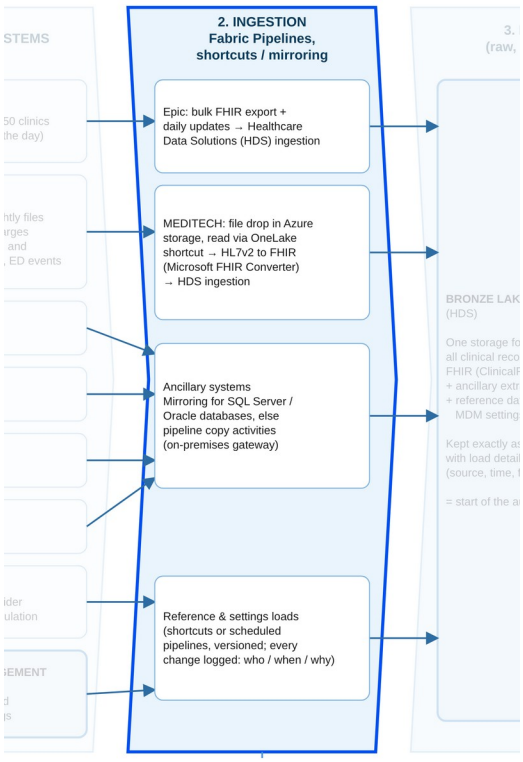


Figure 3. Stage 2: ingestion highlighted in the architecture

Summary

- Epic: a bulk FHIR export loads the history, then daily updates; Healthcare Data Solutions ingests the files
- MEDITECH: files land in Azure storage and are read through a OneLake shortcut; Microsoft's FHIR Converter turns HL7v2 into FHIR, then the same ingestion runs
- Ancillary systems: Mirroring for SQL Server and Oracle databases, otherwise Fabric Pipelines copy activities, through the on-premises gateway
- Reference data and settings arrive through shortcuts or scheduled pipelines; every settings change is logged with who, when and why

KEY DESIGN CHOICE
Converting MEDITECH to FHIR first means one ingestion path and one storage format, and the next acquisition reuses it.

Explanation

Stage 2 brings data into Fabric in three ways: Fabric Pipelines schedule and monitor every load, OneLake shortcuts point to data that already sits in cloud storage without copying it, and Mirroring keeps a near-real-time copy of supported databases.

For Epic, a bulk export pulls the full history as FHIR files, followed by daily updates. Healthcare Data Solutions, Microsoft's healthcare module for Fabric, ingests those files into the bronze layer with its ready-made pipeline.

Healthcare Data Solutions only accepts FHIR files. So for MEDITECH, the nightly HL7v2 files land in Azure storage, which OneLake reads through a shortcut, and Microsoft's open-source FHIR Converter translates each message into FHIR. From that point MEDITECH follows exactly the same ingestion as Epic. That gives us one storage format, and the next acquisition, expected in 12 to 18 months, plugs into the same path. This also allows to change the current feed from MEDITECH to FHIR and more frequent ingestion with minimal changes.

The ancillary systems are kept in sync with Fabric Mirroring where the database supports it (SQL Server and Oracle databases can be mirrored natively), or copied with Fabric Pipelines copy activities, through the on-premises data gateway.

Reference data and master data management settings arrive through OneLake shortcuts or scheduled pipelines. Each approved change to a setting, such as a matching threshold, arrives as a new version with who changed it, when and why, and never overwrites the old value.

Every load writes a record to the monitoring layer: what arrived, how many rows, and whether anything failed.

1. Architecture – Stage 3: bronze

One storage format, kept exactly as received

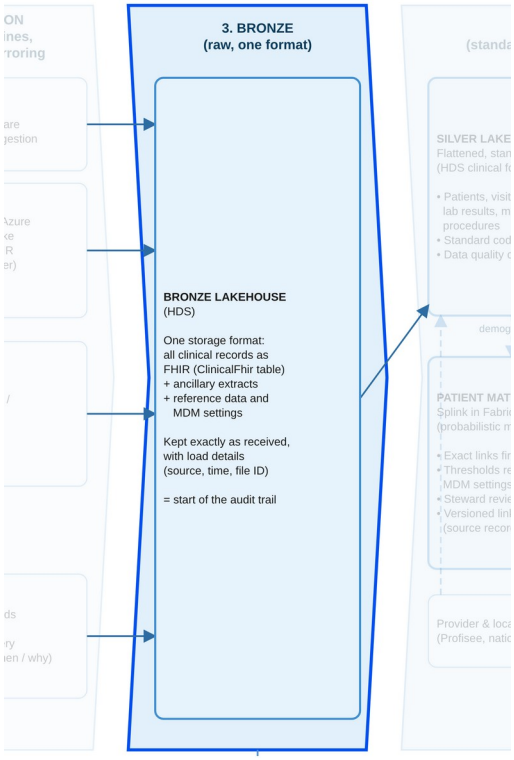


Figure 4. Stage 3: bronze highlighted in the architecture

Summary

- Healthcare Data Solutions stores every Epic and MEDITECH record as FHIR in one bronze table
- Ancillary extracts, reference data and every version of the settings are kept alongside
- Nothing is updated or deleted, so any number traces back to the original message
- Load details (source, time, file ID) recorded for lineage; old versions can be re-run

KEY DESIGN CHOICE
Raw data is never lost, so history can always be recalculated if a rule turns out to be wrong.

Explanation

Stage 3 is the bronze layer, built by Healthcare Data Solutions. Every clinical record from Epic and from MEDITECH, after conversion, is stored in FHIR format in a single bronze table. Before this design, Happy Health had FHIR from Epic and pipe-delimited HL7v2 files from MEDITECH. Now there is one format.

Ancillary extracts, reference data and every version of the master data management settings sit alongside it in the same lakehouse.

Bronze is append-only. Records are never changed or deleted, which is the foundation of the audit trail: any measure result can be traced back through the layers to the message it came from.

Each record carries load details: source system, time received and file ID. Purview uses these to show lineage. Because tables keep their history, we can reprocess from scratch if a mapping or measure rule turns out to be wrong.

Only the platform engineering team can read bronze, because it holds full patient information. Nobody reports directly from it.

1. Architecture – Stage 4: silver

One standard model and one view of each patient

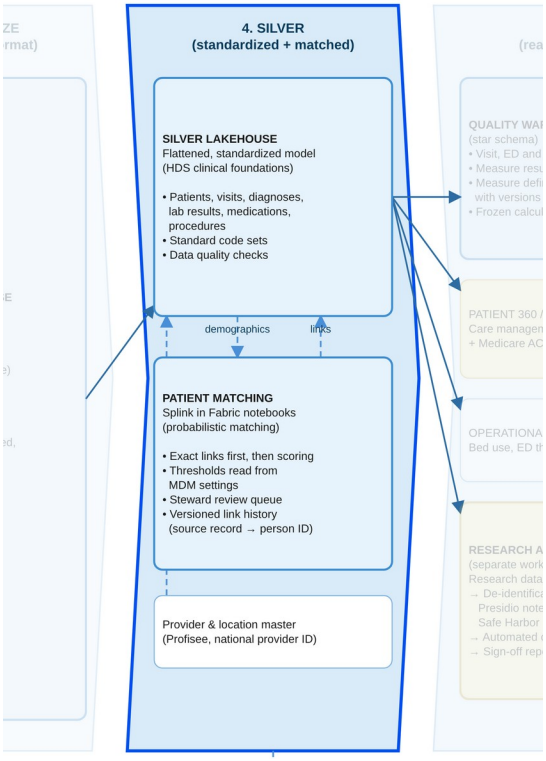


Figure 5. Stage 4: silver highlighted in the architecture

Summary

- Healthcare Data Solutions flattens FHIR into simple tables: patients, visits, diagnoses, results, medications, procedures
- Codes mapped to standard code sets; data quality failures set aside and reported, never silently dropped
- Splink in Fabric notebooks match patients across systems using thresholds from the MDM settings
- A versioned link history records which source record belongs to which person

KEY DESIGN CHOICE
Records are linked, never physically merged, so a wrong merge can be undone without rewriting history.

Explanation

Stage 4 is the silver layer, where data becomes consistent and trustworthy.

Healthcare Data Solutions flattens the nested FHIR records into simple tables for patients, visits, diagnoses, lab results, medications and procedures. Codes are mapped to standard code sets for diagnoses, procedures, lab tests and drugs. Data quality rules check completeness, valid codes and times in the right order. Failing records are set aside and reported, never silently dropped.

Next to the clinical tables sits patient matching. We use Splink, an open-source Python library for probabilistic record matching, running in Fabric notebooks. Shared order and visit numbers create certain links first. Splink then scores the remaining candidate pairs field by field. The thresholds that decide link, review or no link are read from the master data management settings rather than written into code, so data stewards can tune them without a release.

The result is a versioned link history: which source record belongs to which person, when that was true, and when we learned it. Records are linked rather than merged, so a wrong merge can be split cleanly. Provider and location data are mastered in Profisee so that doctors and units line up across both systems.

1. Architecture – Stage 5: gold

Ready-to-use data for each group, all built from silver

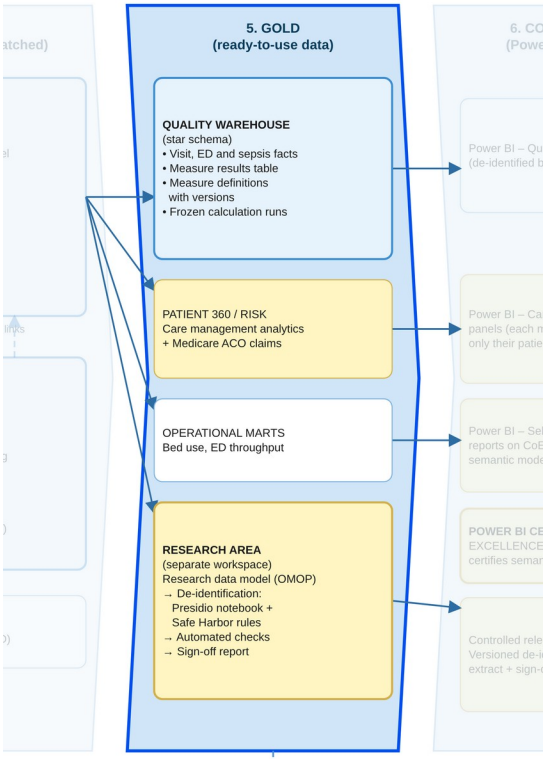


Figure 6. Stage 5: gold highlighted in the architecture

Summary

- Quality warehouse: star schema for the five measures, with frozen calculation runs
- Patient 360 / risk view for the population health team, using matched patients and Medicare claims
- Operational marts for bed use and emergency department throughput, reusing the same visit data
- Research area in its own workspace: research data model, Presidio de-identification, automated checks and a sign-off report

KEY DESIGN CHOICE
The quality warehouse ships first. It funds the program, and every other mart reuses its foundations.

Explanation

Stage 5 is the gold layer: ready-to-use data for each consumer, all built from the same silver data so numbers agree across groups.

The quality warehouse is a star schema, described in deliverable 2. Visit, emergency department and sepsis tables feed a patient-level measure results table, which is summarised into a measure summary table; each calculation run is frozen once submitted.

The patient 360 and risk view serves the population health team. It uses Fabric's care management analytics and the Medicare claims files the ACO (accountable care organization) receives, keyed on the matched person, so a patient's history across every facility is visible in one place.

Operational marts for bed use and emergency department throughput reuse the same visit and emergency department tables. That makes them a cheap quick win.

The research area lives in its own Fabric workspace. Data is first shaped into OMOP, the common research data model most research teams expect. It is then de-identified with two tools working together: Safe Harbor rules for the structured fields (dates cut to the year, ages over 89 grouped, ZIP codes cut to three digits) and Microsoft's open-source Presidio library, run in a Fabric notebook, to find and remove names, numbers and other identifiers in any text. Automated checks run on every release and produce a sign-off report for the privacy officer.

1. Architecture – Stage 6: consumption

The right data for the right person, and no more

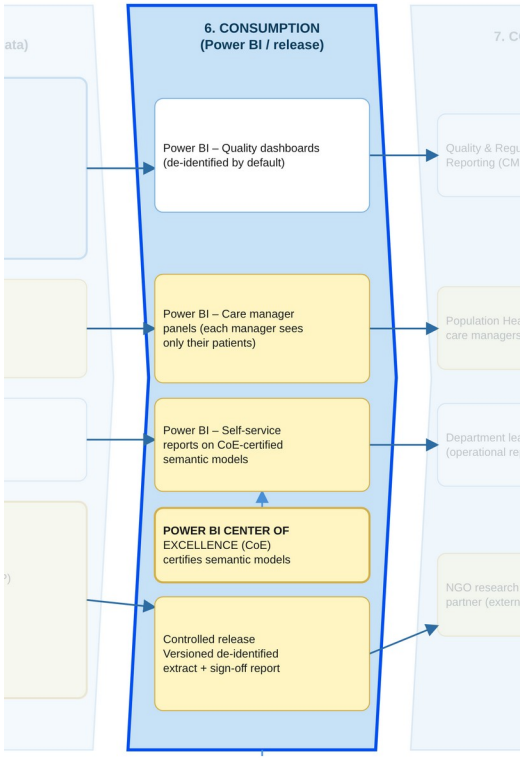


Figure 7. Stage 6: consumption highlighted in the architecture

Summary

- Quality dashboards in Power BI, reading the data directly; de-identified totals by default
- Care manager panels: row-level security shows each manager only their own patients
- Self-service reports on semantic models certified by the Power BI Center of Excellence, so departments don't need an IT ticket
- Research: a versioned de-identified extract, released only after all checks pass

KEY DESIGN CHOICE
Self-service is safe because security lives in the shared models, not in each report.

Explanation

Stage 6 is how each group uses the gold data. It is also where "minimum necessary" access, the HIPAA rule that people see only the data their job needs, becomes visible.

Quality dashboards run in Power BI on Direct Lake models, which read OneLake directly without copying data. Quality analysts see de-identified totals by default. Seeing patient-level detail needs an approved, time-limited grant.

Care managers use Power BI reports with row-level security tied to the panel assignment table, so each manager sees only the patients assigned to them.

Department leaders build their own reports on shared semantic models certified by the Power BI Center of Excellence (see deliverable 5). Because definitions and security live in the shared model, a new dashboard needs no IT ticket and can't accidentally expose patient information.

The NGO gets no platform access. Instead, a controlled release channel publishes a versioned, de-identified extract, only after every automated check passes and the privacy officer signs off.

Every query and export is logged in the audit and monitoring layer.

1. Architecture – Stage 7: consumers

Four groups with four different needs, served in priority order

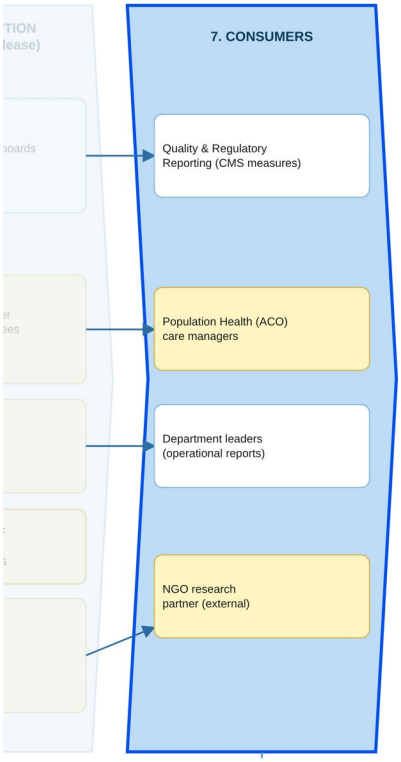


Figure 8. Stage 7: consumers highlighted in the architecture

Summary

- Quality & Regulatory: five automated, auditable CMS measures replacing quarterly spreadsheets, first, within 120 days
- Population health team: one risk view of each patient across every facility, for outreach
- Department leaders: bed use and emergency department dashboards without IT tickets
- NGO research partner: a multi-year extract that is provably de-identified under HIPAA Safe Harbor

KEY DESIGN CHOICE
Order: quality measures first, an operational quick win next, then population health and research once matching is mature.

Explanation

Stage 7 is the four groups waiting on the platform, and the order in which we serve them.

Quality and Regulatory reporting comes first. Today they assemble CMS measures by hand from each system every quarter. The platform replaces that with five automated, auditable measures: 30-day readmissions, SEP-1, emergency department time to provider, OP-18 and OP-22 (left without being seen). The 120-day date is fixed because it keeps the program funded.

The population health team wants one risk view of each patient across every facility they've visited, to target outreach. That depends on mature patient matching and claims linkage, so it follows the quality release.

Department leaders want dashboards such as bed use and emergency department throughput. These reuse the tables built for quality, so a first dashboard can ship within the 120 days as a quick win.

The NGO research partner needs a multi-year, de-identified extract for an outcomes study. The NGO is not part of Happy Health's workforce, so the extract must be provably de-identified. It carries the most compliance risk and has no deadline, so it waits until the research ethics approval, the data use agreement and the privacy review are in place.

1. Architecture – audit & monitoring layer

Every stage reports in; one model answers "is it running, is it right, who touched it?"



Figure 9. Architecture – audit & monitoring layer

Where the information comes from

- Notebook log tables: every notebook run writes rows in and out, rejected rows, errors and the settings version used
- Log Analytics: Spark diagnostics from notebooks
- Workspace monitoring: logs for pipelines, semantic models and Eventhouse, kept 30 days by default

What it produces

- Monitoring Eventhouse: our own audit model of runs, data quality, matching statistics, releases and access, kept for years
- Real-time dashboard: load status by stage and source, late files, error drill-down, audit trail
- Fabric Activator alerts: failed or late loads, data quality breaches, growing review queue, failed de-identification checks

Explanation

The audit and monitoring layer sits under all seven stages and answers three questions in real time: is the process running, is the data right, and who touched what.

It has three sources. First, notebook log tables: every notebook we write follows a logging standard and records a run ID, start and end time, rows read, written and rejected, errors, data quality results and the version of the master data management settings it used. Second, Log Analytics: Fabric's Spark diagnostic emitter sends notebook logs and metrics there. We need this because Fabric's built-in workspace monitoring doesn't cover notebooks today. Third, workspace monitoring itself, a preview feature that collects logs for pipelines, semantic models and Eventhouse into a read-only database, kept for 30 days by default.

All three feed a monitoring Eventhouse holding our own model: run and step tables, data quality checks, matching statistics (automatic links, items waiting for review, no-links), research releases and access events. Because audits look back years, this model keeps its own long-term history rather than relying on the 30-day default.

On top sit a real-time dashboard, showing load status by stage and source, late files, row-count trends and a drill-down from any error to its log message, and Fabric Activator alerts. Examples: a MEDITECH file hasn't arrived by 6 a.m., more than 2% of rows are rejected, the review queue passes its limit, or a de-identification check fails. Alerts go to Teams or email and can restart a pipeline.

2. Data model – the five priority measures

Named in the problem statement, and buildable at all 17 hospitals once the MEDITECH additions arrive

Measure	What it measures and the data it needs
1. 30-day readmission *CMS readmissions program*	Patients readmitted within 30 days of discharge. Needs visit dates, diagnoses and patient matching across sites.
2. SEP-1 sepsis bundle *CMS inpatient reporting*	Timely sepsis care: lactate, blood cultures, antibiotics, fluids. Needs the new MEDITECH pharmacy messages; some items confirmed from notes.
3. ED time to provider *Internal (former CMS measure)*	Minutes from emergency department arrival to first provider. Needs the new MEDITECH ED tracking events.
4. OP-18: ED arrival to departure *CMS outpatient reporting*	Median time in the emergency department for discharged patients. Uses arrival and departure times both systems already send.
5. OP-22: left without being seen *CMS outpatient reporting*	Share of emergency department visits where the patient left before being evaluated. Uses the ED disposition both systems already send.

Explanation

The problem statement names the five measures. All five use the same visit model and can be built at the 12 Epic hospitals and the 5 MEDITECH hospitals.

30-day readmission is tracked internally using the same method as the CMS Hospital Readmissions Reduction Program. SEP-1 is the sepsis care bundle in CMS inpatient quality reporting. It is chart-abstracted, so even with complete data some items, such as the clinician's note of suspected infection, are confirmed by an abstractor; the platform calculates everything else and pre-fills the case. ED time to provider is the former CMS OP-20, tracked with the same method: minutes from arrival to the first provider.

OP-18 is the median time from emergency department arrival to departure for discharged patients. It needs only arrival and departure times, which both Epic and MEDITECH already send. OP-22 is the share of emergency department visits where the patient left before being evaluated. It needs only the ED disposition, which both systems already send, and reuses the same ED visit table as OP-18. CMS replaces OP-18 and OP-22 from January 2028 with its new Emergency Care Access & Timeliness measure, and our ED table already holds the times that measure needs.

All five reuse the same visit model, which keeps the 120-day build realistic.

2. Data model – star schema for quality reporting

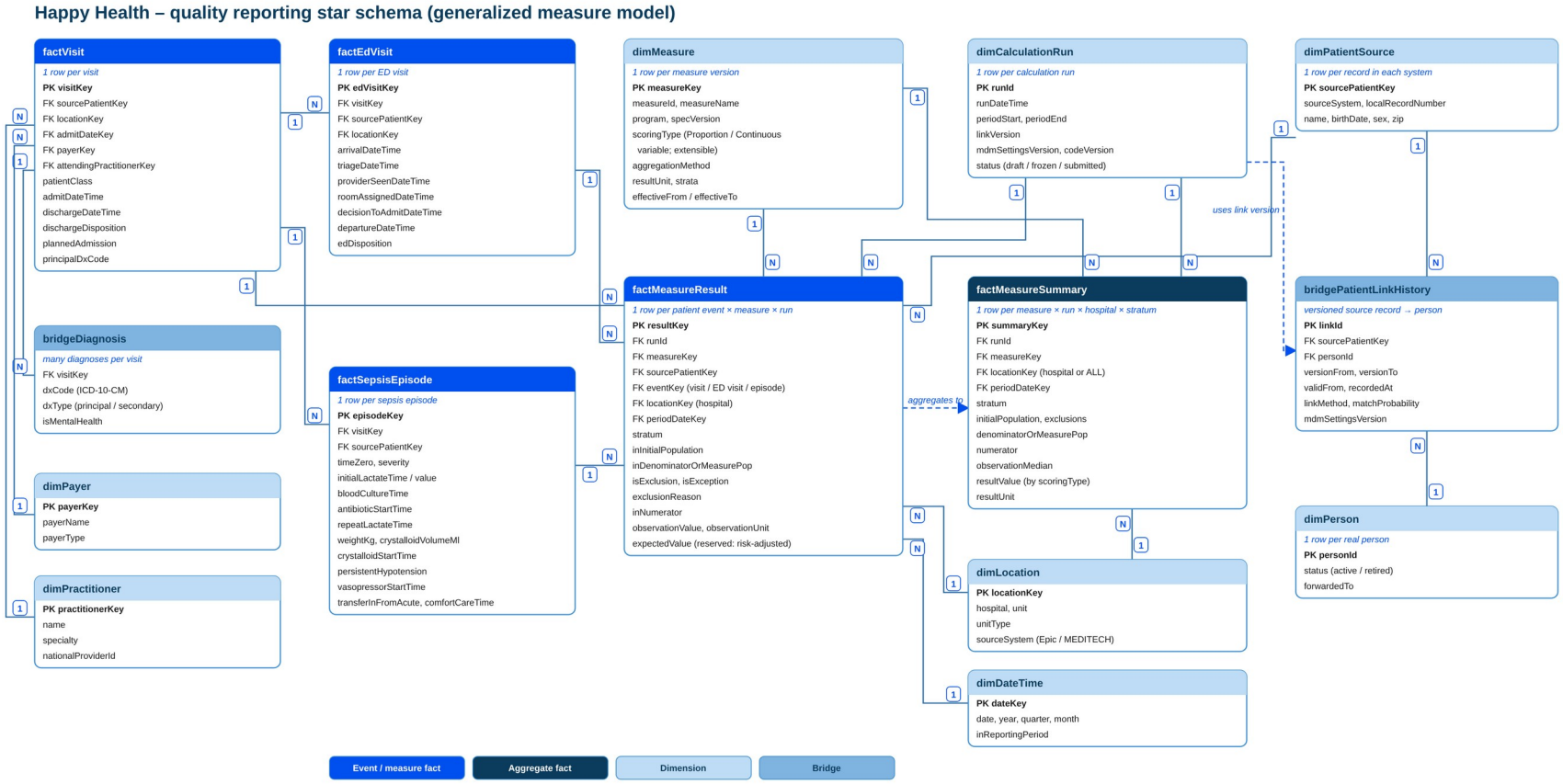


Figure 10. Data model – star schema for quality reporting

Explanation

We chose a star schema rather than one wide table. The five measures share the same descriptive dimensions: date, location, practitioner, payer and measure. Measure definitions change every year and need versions. And auditors need to trace each number to individual rows, which a single flattened table makes hard.

There are two kinds of fact table. Event facts hold what happened, each at its natural grain: one row per visit, one row per emergency department visit (arrival, triage, provider seen, room assigned, decision to admit, departure), one row per sepsis episode (each bundle step's time).

The core fact is the measure results table (factMeasureResult). Its grain is one row per patient event, per measure, per reporting period, per calculation run. Its fields follow CMS's population model: initial population, denominator (or measure population for median-based measures), exclusion and exception with the reason, numerator, an observation value such as minutes, and a stratum. Fields that don't apply to a measure type stay empty, and expectedValue is reserved for a future risk-adjusted measure. dimMeasure records each measure's scoring type, proportion or continuous variable. The factMeasureSummary table then turns the rows into results, one row per measure, run, hospital and stratum, with a single result formula that applies a rate or a median depending on the scoring type. Adding a measure means a new dimMeasure row and logic that emits standard rows, with no new columns or tables.

Example with simulated data. Calculation run R-2026Q3-01, for the July to September 2026 reporting period, produces these factMeasureResult rows for two of the measures:

Measure	Patient record	Event	Denominator	Numerator	Excluded (reason)	Value
OP-18	EPIC-10023	ED-5501	Yes	–	–	142 min
OP-18	EPIC-10577	ED-5502	Yes	–	–	95 min
OP-18	MEDI-88412	ED-7710	Yes	–	–	210 min
OP-18	MEDI-88950	ED-7713	Yes	–	–	168 min
OP-18	EPIC-11002	ED-5507	No	–	Yes (admitted)	305 min
Readmission	EPIC-10023	VIS-3001	Yes	Yes	–	18 days
Readmission	EPIC-10450	VIS-3002	Yes	No	–	–
Readmission	MEDI-88412	VIS-9105	Yes	No	–	–
Readmission	EPIC-10577	VIS-3010	No	–	Yes (died in hospital)	–
Readmission	MEDI-88950	VIS-9110	Yes	No	–	–

How the results are calculated. OP-18 is a median: take the measured values of the rows that are in the denominator and not excluded (95, 142, 168 and 210 minutes), and the median is (142 + 168) ÷ 2 = 155 minutes. The admitted patient (305 minutes) is excluded because OP-18 counts only discharged patients. The readmission rate is numerator ÷ denominator = 1 ÷ 4 = 25%, and the in-hospital death is excluded. Patient EPIC-10023 was readmitted 18 days later at a MEDITECH hospital under record MEDI-90233. That readmission counts only because version 7 of bridgePatientLinkHistory links both records to person P-000417, and dimCalculationRun R-2026Q3-01 records that version. If a later split separated those records, a new run would show 0 ÷ 4 = 0%, while the submitted 25% stays frozen and the difference report explains the change.

On the right is the identity part. Facts point to the source patient record, never directly to the person ID. The patient link history table says which source record belongs to which person, and each calculation run

records the link version and settings version it used. The merges and splits section explains why that matters.

Tables are named in camelCase: fact tables start with fact (factVisit, factMeasureResult), dimensions with dim (dimMeasure, dimPatientSource) and bridge tables with bridge (bridgeDiagnosis, bridgePatientLinkHistory). Every relationship is one-to-many, shown as 1 and N on the diagram. Each dimension row, such as one measure, one location or one calculation run, relates to many factMeasureResult rows. Each event row, such as one emergency department visit, can produce many result rows across measures and runs. One visit has many diagnoses in bridgeDiagnosis. factMeasureSummary is built from factMeasureResult (the dashed arrow), and the run's link version points to bridgePatientLinkHistory. And one dimPatientSource record and one dimPerson each have many rows in bridgePatientLinkHistory over time.

2. Data model – merges and splits without rewriting history

Change the links, never the facts

How it works

- Facts store the source patient record (for example, a MEDITECH record number), never the person ID
- The patient link history records each link with "valid from / valid to" and "recorded at" dates
- A merge or split adds a new link version; the old one is ended, not deleted
- Recalculating creates a new calculation run beside the old one

What an auditor sees

- Submitted results are frozen and stay exactly as filed
- Every result shows which link version and settings version produced it
- A difference report lists any past number that would change today, and why
- Any result can be rebuilt "as of" the day it was submitted

Explanation

This is the heart of deliverable 2: how a merged or split patient identity shows up without silently rewriting history.

The rule is to change the links, never the facts. Fact rows store the source patient record, such as MEDITECH record 12345, which never changes. Which person that record belongs to lives in the patient link history, and every link has two sets of dates: when it was true in the real world, and when we recorded it.

Take an example. Last quarter an Epic record and a MEDITECH record were treated as two people, so a readmission across the two hospitals was missed. This quarter matching links them. We add a new link version; we don't edit the old one. When the measure is recalculated, it runs as a new calculation run using the new version. Last quarter's submitted result stays frozen. A difference report shows that last quarter's readmission rate would now be slightly higher, and why.

A split works the same way. The wrong link is ended, a new person ID is issued, and the affected measures are recalculated as new runs.

For an auditor, every number can be rebuilt as of the day it was submitted, using the link version and settings version recorded on its calculation run.

3. Assumptions and open questions

What we assumed, and what we'd confirm with stakeholders in week one

Assumptions

- MEDITECH nightly files are extended with pharmacy orders and doses, coded diagnoses and procedures, and emergency department tracking events
- Epic's bulk FHIR export can refresh daily within its group limits; otherwise daily changes come from Epic Clarity through mirroring
- MEDITECH HL7 export stays nightly for the 120 days
- Last quarter's spreadsheets can be used for a side-by-side check
- A named requirements owner has the authority to answer questions and confirm changes

Open questions for week one

- Can the MEDITECH interface team deliver the MEDITECH additions in time, and at what cost?
- Who owns patient identity and the matching thresholds today?
- Where do care manager panel assignments come from?
- Does the research study need exact dates? Safe Harbor keeps only the year
- Why not use MEDITECH's FHIR interface to get near-real-time data from the newly acquired hospitals?
- What platform does each ancillary system run on? Are any of them SQL Server or Oracle databases that Fabric can mirror natively?
- Do any sources carry Social Security numbers (full or last four digits), and will the privacy officer allow their use in matching?

Explanation

The problem statement is deliberately underspecified, so we state what we assumed and what we'd confirm in week one rather than guess.

Assumptions. First, the MEDITECH nightly files will be extended with three additions: pharmacy orders and doses given, coded diagnoses and procedures, and emergency department tracking events. Without them, SEP-1 and time to provider can't be fully calculated at the five MEDITECH hospitals. Our fallback is for abstractors to fill a structured template that is loaded into the pipeline. Second, Epic's bulk FHIR export can refresh daily within its limits (group-based, no changed-since filter, once per group per day by default); otherwise daily changes come from Epic's Clarity reporting database through mirroring. Third, MEDITECH stays on nightly files for the full 120 days. Fourth, last quarter's spreadsheets can serve as the baseline for a side-by-side check. Fifth, a named requirements owner has the authority to answer questions and confirm changes, so decisions don't stall.

Open questions. Can the MEDITECH interface team deliver the MEDITECH additions in time, and what does the contract allow? Who owns patient identity today, usually Health Information Management, and who should approve threshold changes? Where do care manager panel assignments live? Do any sources carry Social Security numbers, full or only the last four digits, and will the privacy officer allow them to be used for matching? The problem statement's source reference doesn't list them in any source. For research, which data items and how precise do dates need to be? Safe Harbor removes all dates except the year, which may

not suit an outcomes study. If not, Expert Determination is the alternative to raise. Finally: who handles the research ethics approval and the data use agreement?

Two more questions shape ingestion. First, why not use MEDITECH's FHIR interface for near-real-time data from the newly acquired hospitals? MEDITECH Expanse offers FHIR APIs, while older MEDITECH platforms may not. If the acquired hospitals run Expanse and the APIs can be licensed and enabled, they would give fresher data in the same format as Epic and remove the HL7v2 conversion step. We didn't plan on it because the problem statement says the hospitals are batch-only with no funded interface project, but because everything is converted to FHIR, switching later is a small change. Second, what platform does each ancillary system run on? Fabric mirroring is generally available for SQL Server 2016 to 2025 and supports Oracle 10 and later (confirm whether Oracle mirroring is still in preview), both through the on-premises gateway. Mirroring keeps a near-real-time copy in OneLake with no pipeline to maintain, so the answer decides between mirroring and scheduled pipeline copies for each system.

4. Patient matching – logical steps of a match and a split

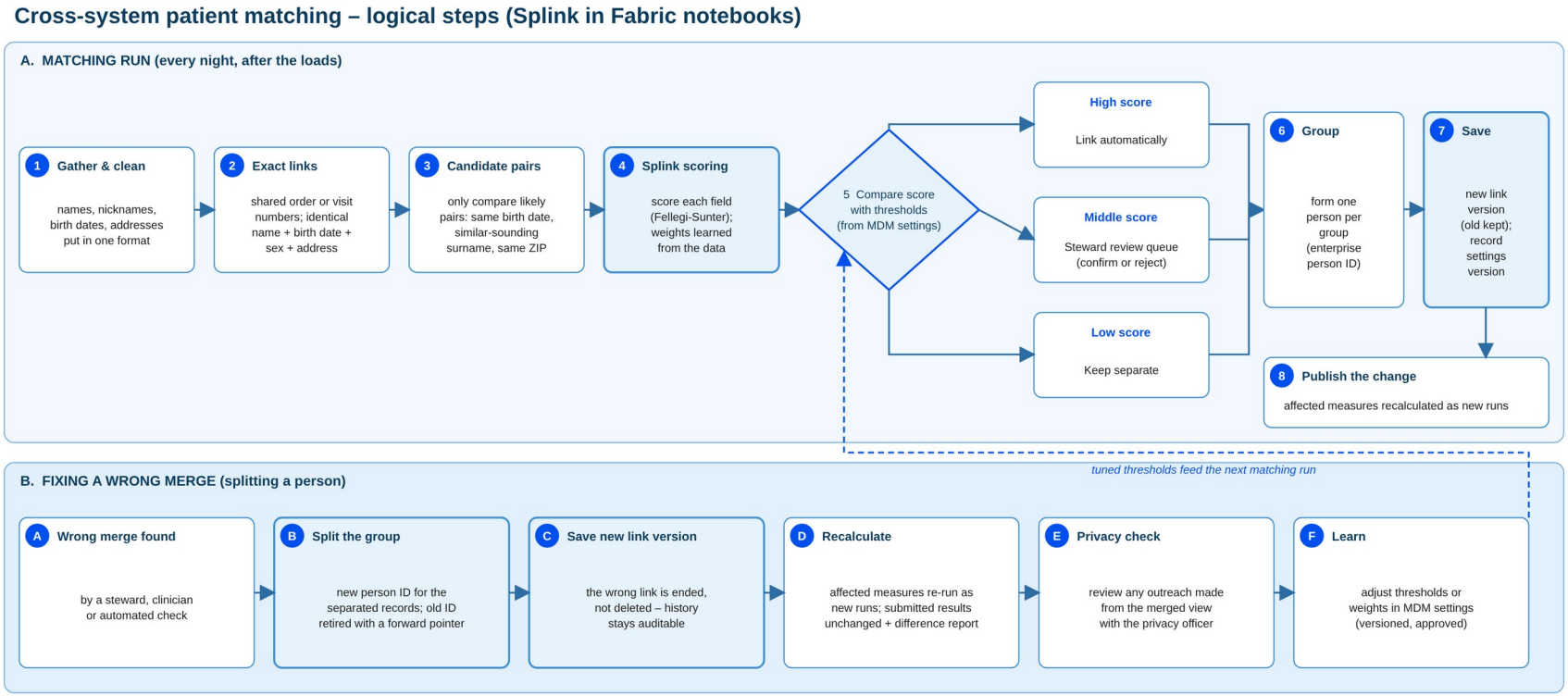


Figure 11. Patient matching – logical steps of a match and a split

Explanation

This diagram shows the logical steps of cross-system patient matching, which runs every night after the loads.

Step 1 cleans the data: names, nicknames, dates of birth and addresses are put in one format.

Step 2 creates certain links first: shared order or visit numbers between the lab system and the electronic record, and identical name, birth date, sex and address.

Step 3 limits comparisons to likely pairs, such as the same birth date, a similar-sounding surname or the same ZIP code, so we don't compare every record with every other.

Step 4 is the Splink scoring. Splink applies the Fellegi-Sunter method: each field contributes a weight depending on how well it agrees, and the weights are learned from Happy Health's own data.

Step 5 compares the total score with two thresholds held in the master data management settings. High scores link automatically. Middle scores go to a review queue worked by Health Information Management data stewards. Low scores stay separate.

Steps 6 to 8 group linked records into one person with a permanent enterprise ID, save a new link version (the old one is kept), and publish the change so the affected measures are recalculated.

The lower lane shows a split, when two merged records turn out to be two people. The group is split, a new ID is issued, the old ID is retired with a forwarding pointer, measures are recalculated as new runs, any outreach based on the wrong merge is reviewed with the privacy officer, and the thresholds are tuned. Those tuned thresholds feed the next matching run.

4. Patient matching – Splink approach and handling uncertainty

Probabilistic matching in Fabric notebooks, with people deciding the grey zone

How Splink is used

- Runs in a Fabric notebook on Spark, at Happy Health's volumes
- Compares name, birth date, sex, address and phone with fuzzy similarity levels
- Learns field weights from Happy Health's own data, then checks them against a hand-labelled sample
- Reports precision and recall, not just a "match rate"; version pinned for repeatable results
- Social Security number, if a source has it: one weighted field, placeholders filtered out, never decisive on its own

Uncertain matches and wrong merges

- Two thresholds from MDM settings: link, review or keep separate
- A false merge is worse than a missed link, so thresholds start cautious
- Claims subscriber ID is never used alone, because it can belong to a family member
- Wrong merges are split cleanly: new ID, old ID forwarded, measures re-run, privacy review

Explanation

We use Splink, an open-source Python library for probabilistic record matching, in a Fabric notebook on the Spark engine, which handles Happy Health's volumes comfortably.

Each pair of candidate records is compared field by field: name with fuzzy similarity levels, so "Jon" and "John" score partly, plus birth date, sex, address and phone. Splink learns how much each level of agreement should count by estimating the weights from Happy Health's own data, using a statistical method called expectation-maximization. We then check the results against a hand-labelled sample of Epic and MEDITECH pairs, and report precision (how many links are right) and recall (how many true links we found), not a vague "match rate". The Splink version is pinned so results are repeatable. A new major version was released in September 2026, so we would test it before upgrading.

Handling uncertainty: two thresholds, stored in the master data management settings, split pairs into link, review and keep separate. The two kinds of error have different costs. A false merge mixes two people's records, a safety and privacy problem. A missed link undercounts readmissions. So thresholds start cautious and are tuned with stewards.

Claims are a special case: the insurance subscriber ID can belong to a parent or spouse, so it's never used alone.

The problem statement's source reference doesn't list Social Security numbers in any source, so the design doesn't depend on them. If a source does carry them, we add the number as one weighted field in Splink, not as a deciding key. In healthcare it's often missing (newborns, some immigrants, patients who decline, unidentified emergency patients), often wrong (placeholders such as 000-00-0000, typos, a parent's number on a child's record) and sometimes shared through identity theft. Splink handles this well: a missing value adds no evidence, a one-digit difference counts as partial agreement, and common placeholder values are filtered out or given little weight. An SSN match with a different name and birth date still goes to review. Because it's highly sensitive, we bring in only what's needed, possibly the last four digits, keep it in restricted

columns excluded from all reports and the research extract, and agree with the privacy officer whether to tokenize it, which is safer but allows exact matches only.

When a merge turns out wrong, the person is split cleanly, as shown in the matching steps diagram. A commercial matching product can replace the matching engine later behind the same link history, without changing the data model.

4. Patient matching – sample of the records the process produces

What the nightly matching run writes, shown with records from the synthetic data workbook

Scored candidate pairs

Matching run · settings	Record A (Epic)	Record B (MEDITECH)	How the fields compare	Probability	Outcome
1 May 2026 · MDM-1	EPIC-10015 Joseph Thomas, 22 Feb 1946, M, 33399	MEDI-88015 Joseph Thomas, 22 Feb 1946, M, 33399	All fields identical	1.00	Linked by the exact rule (step 2) to person P-0015; link version 1
1 May 2026 · MDM-1	EPIC-10012 Barbara Gonzales, 22 Jan 1964, F, 33247	MEDI-88012 Barbara Gonzalez, 22 Jan 1965, F, 33401	Surname one letter apart; birth year one apart; same phone; different ZIP	0.97	Above the 0.95 auto-link threshold, linked to P-0012 (later found wrong)
1 May 2026 · MDM-1	EPIC-10005 John Jones, 21 Jun 1956, M, 33963	MEDI-88005 John Jones, birth date not yet sent, M, 33963	Very common name, so it carries little weight; no birth date; different phone	0.12	Below 0.50, kept separate as new person P-0041
29 Sep 2026 · MDM-2	EPIC-10005 John Jones, 21 Jun 1956, M, 33963	MEDI-88005 John Jones, 21 Jun 1956, M, 33963	Birth date now received and identical; phone still different; weights retrained	0.81	Review band (0.50–0.95); steward confirmed; P-0041 retired into P-0005; link version 7
14 Oct 2026 · MDM-3	EPIC-10012 Barbara Gonzales, 22 Jan 1964, F, 33247	MEDI-88012 Barbara Gonzalez, 22 Jan 1965, F, 33401	Same fields; shared phone now weighs less and a different birth year counts against	0.38	Steward confirmed the split; new person P-0042; link version 8

Steward review queue item

Review queue item	Value
reviewId	RV-0193
Pair	EPIC-10005 ↔ MEDI-88005
Match probability	0.81 (review band, MDM-2)
Field evidence	Name and birth date identical, ZIP identical, phone different
Assigned to	Health Information Management data steward
Decision	Same person – link
Reason	Same address and next of kin in both charts
Decided at	29 Sep 2026 16:40; published in the 30 Sep run as link version 7

Link history rows written (bridgePatientLinkHistory)

linkId	Source record	Person	version From	version To	Valid from	Recorded at	Method	Prob.	Settings
LK041	MEDI-88015	P-0015	1	999	01 May 2026	01 May 2026 02:00	Exact	1.00	MDM-1
LK042	MEDI-88005	P-0041	1	6	01 May 2026	01 May 2026 02:00	No match	0.12	MDM-1
LK043	MEDI-88005	P-0005	7	999	28 Jul 2026	30 Sep 2026 03:00	Steward	0.81	MDM-2
LK044	MEDI-88012	P-0012	1	7	01 May 2026	01 May 2026 02:00	Scored	0.97	MDM-1
LK045	MEDI-88012	P-0042	8	999	20 Aug 2026	15 Oct 2026 03:00	Steward (split)	0.00	MDM-3

Enterprise person records (dimPerson)

personId	Status	Forwarded to	Source records in the latest link version
P-0005	Active		EPIC-10005, and MEDI-88005 from link version 7
P-0012	Active		EPIC-10012 only, from link version 8
P-0041	Retired	P-0005	Retired in link version 7 when MEDI-88005 was matched to P-0005
P-0042	Active		Created in link version 8 when MEDI-88012 was split from P-0012

Explanation

These records come from the synthetic data workbook, so they line up with the measure examples. The MEDI-88005 link is why the readmission at the MEDITECH hospital counts in run R-2026Q3-01 (14.3%), and the MEDI-88012 split is why the re-run R-2026Q3-02 shows 11.9%.

Each nightly run writes three kinds of record. First, scored candidate pairs: every likely pair found in step 3, how each field compares, the match probability and the outcome. They're kept, so anyone can see why two records were or weren't linked. Second, review queue items for pairs in the middle band, with the steward's decision and reason. Third, whenever something changes, a new link version: new rows in bridgePatientLinkHistory and, where needed, new or retired person IDs in dimPerson.

The settings versions come from the master data management source. MDM-1 used starting weights, linking automatically at 0.95 or above and sending pairs from 0.50 to the review queue. MDM-2 retrained the field weights on Happy Health's own data. MDM-3, approved after the wrong merge, gave a shared phone number less weight, because family members often share one, and made a different birth year count against a match, so the Gonzales and Gonzalez pair now scores 0.38.

How to read the link history: versionFrom and versionTo give the link versions a row belongs to, and 999 means it is still current. MEDI-88005 belonged to P-0041 in versions 1 to 6 and to P-0005 from version 7. MEDI-88012 belonged to P-0012 in versions 1 to 7 and to the new person P-0042 from version 8. Nothing is overwritten: old rows are closed and new rows are added. Valid from is when the link became true in the real world, such as the MEDITECH admission on 28 July; recorded at is when the matching run learned it.

5. Governance – access scoped by role and panel

Enforced by the platform, provable to auditors



Figure 12. Governance – access scoped by role and panel

Who sees what

- Care managers: patient detail, but only for their assigned panel
- Quality analysts: de-identified totals; patient detail only with an approved, time-limited grant
- Department leaders: operational totals; engineers: emergency access only, logged

How it's enforced

- Purview sensitivity labels on every patient-information column drive masking automatically
- Entra ID groups, workspace roles, and row- and column-level security in OneLake
- Settings changes approved by stewards; quarterly access reviews; every query logged

Explanation

The electric-blue band runs under every stage because governance isn't a step you pass through once. It applies everywhere.

Access follows the HIPAA minimum-necessary rule. Care managers see patient detail, but row-level security limits them to their assigned panel. Quality analysts see de-identified totals by default. Patient-level detail needs an approved grant that expires. Department leaders see operational totals. Engineers get emergency access only, and every use is logged.

Enforcement is built into the platform, not left to each report. Microsoft Purview labels every column that holds patient information, and those labels drive masking and access policies automatically. Entra ID groups and Fabric workspace roles control who can open what. OneLake security adds row-level and column-level access.

Three things make it provable. Every query and export is logged in the audit and monitoring layer. Access is reviewed every quarter. And changes to the master data management settings, such as raising a matching threshold, go through steward approval and are versioned, because a threshold change can move a measure result and has to be explainable.

Underneath: encryption, keys held in a key vault, private networking, a signed HIPAA business associate agreement with Microsoft, and version-controlled deployments from development to test to production.

5. Governance – proving the research extract is de-identified

Safe Harbor rules + Presidio, with evidence produced on every release

How the data is de-identified

- Start from an approved list of columns; nothing else can leave
- Safe Harbor rules for structured fields: dates cut to the year, ages over 89 grouped, ZIP cut to 3 digits (000 for small areas)
- Presidio, in a Fabric notebook, finds and removes names, numbers, phones and emails in any text
- Random study IDs replace record numbers and are never derived from them

Evidence on every release

- Automated checks: no identifier patterns found, date and ZIP rules met, column list unchanged
- A failed check blocks the release and raises an Activator alert
- Sign-off report: check results, lineage and tool versions, signed by the privacy officer
- Released only through the separate research workspace's controlled channel

Explanation

The NGO is not part of Happy Health's workforce, so the extract has to be provably de-identified under HIPAA Safe Harbor, not just access-controlled. Safe Harbor means removing 18 kinds of identifier and having no actual knowledge that the remaining data could identify someone.

How it's done. We start from an approved list of columns, and nothing outside it can leave. Structured fields follow Safe Harbor rules: all dates reduced to the year, ages over 89 grouped as 90 or older, ZIP codes cut to three digits and set to 000 where that three-digit area has 20,000 people or fewer, using Census data. Microsoft's open-source Presidio library, run in a Fabric notebook, scans any text for names, record numbers, phone numbers, emails and similar identifiers and removes them. We add custom patterns for Happy Health's own record-number formats. Study IDs are random and never derived from record numbers.

How it's proven. Every release runs automated checks: Presidio and pattern scanners find no identifiers, date and ZIP rules are met, and the column list matches the approved version. A failed check blocks the release and raises an Activator alert. A sign-off report records the check results, lineage and tool versions, and the privacy officer signs it. The extract leaves only through the research workspace's controlled release.

5. Governance – Power BI Center of Excellence: certifying semantic models

Self-service is safe because every shared model is reviewed before people build on it

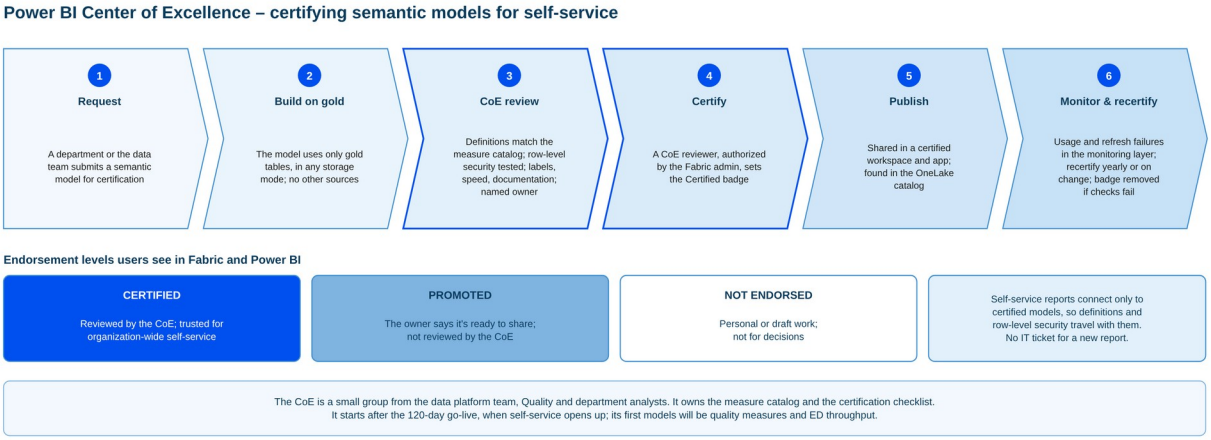


Figure 13. Power BI Center of Excellence – certifying semantic models

Certification steps

- Request: a department or the data team submits a semantic model for certification
- Build on gold: the model uses only gold tables, whatever its storage mode (Direct Lake, import or DirectQuery), with no other sources
- CoE review: definitions match the measure catalog; row-level security tested with test users; sensitivity labels, speed and documentation checked; a named owner
- Certify: a CoE reviewer, authorized by the Fabric administrator, sets the Certified badge
- Publish: shared in a certified workspace and app, and found in the OneLake catalog
- Monitor and recertify: usage and refresh failures tracked in the monitoring layer; recertified yearly or when the model changes; badge removed if checks fail

Explanation

Self-service reporting only stays safe if the shared semantic models people build on are correct and secure. The Power BI Center of Excellence (CoE) is the process, and the small team, that makes sure of that. It brings together the data platform team, the Quality team and analysts from the departments, and it owns two things: the measure catalog, meaning the agreed definition of every measure, and the certification checklist.

A model goes through six steps. Anyone can request certification. The model must be built only on gold tables. The storage mode doesn't matter (Direct Lake, import or DirectQuery), but no other sources, side files or private extracts are allowed. The CoE then reviews it: definitions match the measure catalog, row-level security is tested with test users for each role, sensitivity labels are applied, refresh and query speed are acceptable, documentation exists, and a business owner is named. If it passes, a CoE reviewer sets the Certified badge. In Fabric, only people the Fabric administrator authorizes can certify, so the badge means something. The model is then published in a certified workspace and app, and users find it in the OneLake catalog.

Certification isn't permanent. The monitoring layer tracks usage and refresh failures for every certified model. Models are recertified every year or whenever they change, and the badge is removed if a check fails or the owner leaves.

Users see three levels. Certified models are reviewed by the CoE and trusted organization-wide. Promoted models are ones the owner considers ready to share but the CoE hasn't reviewed. Everything else is personal or draft work. Self-service reports connect only to certified models, so definitions and security travel with them and a new report needs no IT ticket. The CoE starts after the 120-day go-live, when self-service opens up, and its first models will be quality measures and emergency department throughput.

6. 120-day plan – four phases to five live measures

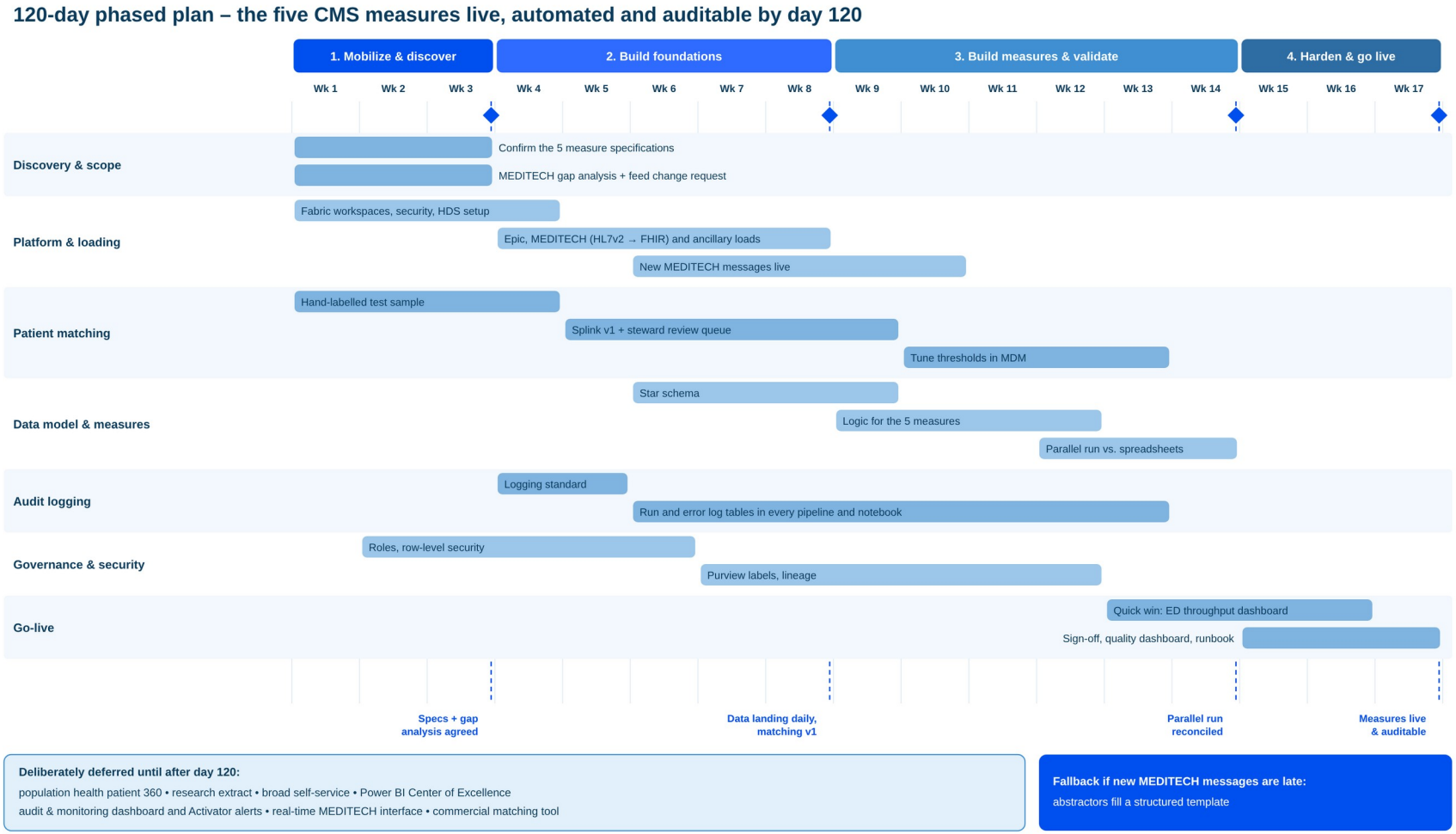


Figure 14. 120-day plan – four phases to five live measures

Explanation

The plan has four phases across 17 weeks (120 days).

Phase 1, weeks 1 to 3: mobilize and discover. Confirm the exact specifications of the five measures with the quality team, and complete the MEDITECH gap analysis and change request so work on the feed additions can start. Set up the Fabric workspaces, security and Healthcare Data Solutions, and start a hand-labelled sample for testing matching.

Phase 2, weeks 4 to 8: build foundations. Daily loads from the 12 Epic hospitals, the 5 MEDITECH hospitals (converted from HL7v2 to FHIR) and the ancillary systems. Splink matching version 1 with the steward review queue. The logging standard, so every notebook reports to the monitoring layer from day one. Roles and row-level security.

Phase 3, weeks 9 to 14: build and validate the measures. Star schema and the logic for the five measures. Most important is a side-by-side run against last quarter's spreadsheets in weeks 12 to 14, where every difference is explained with clinical experts.

Phase 4, weeks 15 to 17: harden and go live. Sign-off, the quality dashboard and the runbook, plus a quick win for department leaders, an emergency department throughput dashboard built from the same tables.

Milestones: specifications and gap analysis agreed by week 3, data landing daily with matching version 1 by week 8, side-by-side check reconciled by week 14, measures live and auditable by week 17.

6. 120-day plan – what ships, what waits, and why

The deadline funds the program, so everything in the 120 days serves the five measures

Ships within 120 days

- Five CMS measures, automated and auditable, checked against last quarter
- Daily loads from all sources in one format, with patient matching version 1
- Logging standard: every pipeline and notebook writes run and error records, so the audit trail exists from day one
- Quick win: emergency department throughput dashboard for department leaders

Deliberately deferred

- Population health patient 360: needs mature matching and claims linkage
- Research extract: highest compliance risk, no deadline, needs ethics approval and a data use agreement
- Broad self-service, a real-time MEDITECH interface, and a commercial matching product decision
- Power BI Center of Excellence, which starts when self-service opens up after go-live
- Audit & monitoring dashboard and Activator alerts, built on the log tables collected from day one

Explanation

Why this split is right. The 120-day deadline is what funds the program, so everything inside the 120 days serves the five measures, and the measures need exactly the foundations every other group will reuse: loading, one storage format, patient matching, the visit model and the audit trail.

What ships: the five measures, automated and auditable, reconciled against last quarter's manual numbers. Daily loads from every source in one format. Patient matching version 1 with a review queue. The logging standard, so every run is recorded for the audit trail from day one. And one quick win for department leaders, an emergency department throughput dashboard that costs little because it reuses the measure tables.

What waits, and why. The population health patient 360 needs mature matching and the claims linkage, which improve after go-live. The research extract carries the most compliance risk, has no deadline and needs ethics approval, a data use agreement and privacy review first. Broad self-service, a real-time MEDITECH interface and the decision on a commercial matching product are all worth doing, but none is needed for the measures. The Power BI Center of Excellence also waits: until self-service opens up, the data team publishes the quality and emergency department dashboards itself, so there is nothing to certify yet. The audit and monitoring dashboard and its Activator alerts wait too. The log tables they read are collected from day one, so nothing is lost, and the team checks runs directly in the meantime.

If the MEDITECH feed additions arrive late, the measures still go live: abstractors fill a structured template for the missing items, which is loaded into the pipeline and stays auditable, and it is retired once the feeds arrive.

6. 120-day plan – bonus: ED throughput dashboard

A quick win for department leaders, built from the tables the measures already use

What it shows, by hospital and by week or day:

View	Calculated from	Why it matters
ED arrivals by hour and day of week	arrivalDateTime	Staffing and peak demand
Median arrival → provider (minutes)	arrival, providerSeen	Same logic as the ED time-to-provider measure
Median arrival → departure, discharged patients	arrival, departure, disposition	Same logic as OP-18
Median ED time, admitted patients	arrival, departure	The longer stays OP-18 leaves out
Boarding time (decision to admit → departure)	decisionToAdmit, departure	Shows inpatient bed pressure
Left without being seen rate	edDisposition	Same logic as OP-22
Disposition mix (discharged, admitted, transferred, LWBS, AMA)	edDisposition	Overall flow
ECAT readiness: share of visits waiting > 1 h for a room, boarding > 4 h, or in the ED > 8 h	roomAssigned, decisionToAdmit, departure	Previews the CMS measure that replaces OP-18 and OP-22 in 2028

Ground rules

- Totals and medians only, no patient names: department leaders don't need patient detail, so it follows the minimum-necessary rule
- Built and published by the data team, not self-service; the Center of Excellence and certified self-service come after day 120
- Freshness: Epic hospitals update during the day; the five MEDITECH hospitals show yesterday's data until a real-time MEDITECH feed exists
- MEDITECH dependency: provider-seen, room and boarding times at the MEDITECH hospitals need the new ED tracking events; until then those hospitals show volumes, arrival-to-departure and LWBS only, flagged as partial

What it doesn't include

- A live tracking board of patients currently in the ED (Epic already provides one)
- Staffing or cost data
- Inpatient bed occupancy, the other half of the operational marts, which comes after the 120 days

Explanation

Why it's a cheap quick win: every number reuses factEdVisit and the logic already built for ED time to provider, OP-18 and OP-22. The only new work is the report itself, a few days of effort in weeks 13 to 16. It also previews the Emergency Care Access & Timeliness measure that CMS makes mandatory in 2028, so department leaders see those numbers well before they are reported.

7. Risks and mitigations

Risk	Mitigation
1. MEDITECH feed additions arrive late or incomplete	Request in week 1; check every field as it arrives; structured abstraction template as a fallback
2. Patient matching errors: false merges harm safety and privacy; missed links undercount readmissions	Cautious thresholds in MDM settings, steward review queue, versioned links, precision and recall reported on every matching run
3. Measure logic doesn't match CMS specifications or last quarter's numbers	Versioned measure definitions, side-by-side run against spreadsheets, sign-off by clinical experts
4. Healthcare Data Solutions is now customer-maintained code	Treat it as owned accelerator code under version control; keep conversions and models in open formats
5. Next acquisition (12–18 months) brings another record system	Convert every source to FHIR at ingestion; matching built for any number of sources; reusable loading pattern

Explanation

Five risks we'd flag in our own design.

One: the MEDITECH feed additions arrive late or incomplete. They're needed for SEP-1 and time to provider at five hospitals. We raise the request in week 1, check every field as it arrives, and keep a structured abstraction template ready as a fallback, so the measures still go live.

Two: patient matching errors. A false merge mixes two people's records, a safety and privacy problem. A missed link undercounts readmissions. Mitigations: cautious thresholds held in the master data management settings, a steward review queue, versioned links that make any split clean, and precision and recall reported on every matching run.

Three: measure logic that doesn't match the CMS specification or last quarter's manual numbers. We version every measure definition, run side by side against the spreadsheets, and have clinical experts sign off on every difference.

Four: Healthcare Data Solutions is now delivered as code that Happy Health maintains, following Microsoft's change in delivery model. We treat it as our own accelerator code, keep it under version control, and keep the conversions and models in open formats.

Five: the next acquisition, expected in 12 to 18 months, may bring a third record system. Because every source is converted to FHIR at ingestion, matching handles any number of sources, and loading follows a reusable pattern, a new system becomes configuration work rather than a redesign.

8. Platform sizing – Fabric capacity and cost

One F64 for production, Spark billed on demand, smaller paused capacities for UAT and dev

Recommended capacities by environment:

Environment	Capacity	Billing	Why
Production: Power BI and gold	F64	1-year reservation after 1–2 months on pay-as-you-go	Free viewers for about 500 report users; Direct Lake up to 25 GB and 1.5 billion rows per table
Production: loads and notebooks	Autoscale billing for Spark, capped at 64–128 CU	Pay per use	Nightly jobs don't take capacity from reports
UAT	F32; F64 for the parallel run (weeks 12–14)	Pay-as-you-go, paused outside test windows	Real data with production security; performance tested at the production size
Dev	F8	Pay-as-you-go, paused nights and weekends	Synthetic data only

Assumptions behind the sizing

- 17 hospitals and about 150 clinics: about 2.5 million patients, 200,000 admissions, 750,000 ED visits and 3 million clinic visits a year
- About 500 million lab and vital-sign results a year (2.5 billion over five years) and 5–10 TB of raw FHIR in bronze
- About 500 report viewers and 40 report creators
- Rough monthly cost: F64 reserved about $5,000; on-demand Spark about $300; UAT about $1,700; dev about $525; storage about $200; Pro licenses for creators about $560; about $8,300 in total
- Confirmed with the Fabric Capacity Metrics app during the first one to two months, before the reservation is bought

Explanation

The F64 decision rests mainly on report viewers. From F64 up, people who only view reports need a free license; below it every viewer needs a Power BI Pro license at about $14 a month, which for about 500 viewers costs more than the capacity itself. F64 also raises the Direct Lake limits to 25 GB of memory and 1.5 billion rows per table.

The quality models are small, so they fit easily. The full five-year lab-results history, about 2.5 billion rows, would not fit in one Direct Lake model on F64; it stays in the lakehouse, and models read summarized tables. A model that truly needed every row would need F128 or an import model with aggregations.

Nightly loads, patient matching and the measure notebooks run on Spark with autoscale billing. They are charged per use and don't consume the F64, so heavy background work can't slow down reports during the day. The one-time history load is billed the same way, with no need to scale up the capacity.

UAT runs on F32 and is paused outside test windows, then raised to F64 for the parallel run so performance is tested at the production size. Dev uses F8 with synthetic data only. Before buying the one-year reservation, the team watches the Capacity Metrics app for one to two months on pay-as-you-go.